

Part A - Theory Questions and Answers

1. What is Regularization in Machine Learning? Why is it needed?

Regularization is a technique used to reduce overfitting in machine learning models by adding a penalty term to the loss function. It helps the model generalize better on unseen data by preventing it from learning noise from the training dataset.

2. Differentiate between Ridge Regression (L2) and Lasso Regression (L1).

Ridge Regression uses L2 regularization and adds the squared values of coefficients as a penalty term. It reduces coefficient values but does not make them exactly zero. Lasso Regression uses L1 regularization and adds the absolute values of coefficients as a penalty term. It can reduce some coefficients to exactly zero, making it useful for feature selection.

3. What is Cross Validation and why is it important?

Cross Validation is a technique used to evaluate machine learning models by splitting the dataset into multiple training and validation sets. It provides a more reliable estimate of model performance and helps reduce the risk of overfitting.

4. Explain K-Fold, Stratified K-Fold, LOOCV and Time Series Split.

K-Fold Cross Validation: The dataset is divided into K equal parts. Each part is used once as validation data while the remaining parts are used for training.

Stratified K-Fold: Similar to K-Fold, but maintains the distribution of target values across all folds.

Leave-One-Out Cross Validation (LOOCV): Each observation is used once as validation data while the remaining observations are used for training.

Time Series Split: Used for sequential data where future observations should not be used to predict past observations. It preserves the chronological order of the data.

5. Why are tree-based models less sensitive to feature scaling?

Tree-based models such as Decision Trees and Random Forests split data based on feature values rather than distances. Since the splitting process is not affected by the scale of the features, these models generally do not require feature scaling.